

Yiwen Lu

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RESEARCH INTERESTS

My research centers on **self-improving and trustworthy LLM agents**: how agents acquire knowledge, translate it into reliable decisions, and improve through feedback and interaction.

- **Agentic Post-Training**: Improving agent decision-making through supervision, reinforcement learning, and environment feedback.
- **Experience Memory and Continual Adaptation**: Retaining and transferring knowledge across tasks and model executors.
- **Reliable Long-Horizon Agents**: Failure analysis, robust tool use, and evaluation of multi-step agent behavior.

EDUCATION

University of Notre Dame, Notre Dame, IN Aug. 2025 – Present
Ph.D. student in Electrical Engineering; qualifier passed Aug. 2026; M.S. expected Dec. 2026.
Beijing Jiaotong University, Beijing, China Sept. 2021 – Jul. 2025
B.S. in Computer Science.

PUBLICATIONS AND MANUSCRIPTS

1. **Know2Act: What Knowledge Reaches a Tool-Using Agent’s Actions?**
Yiwen Lu *Manuscript, 2026* [PDF](#) [Code](#) [Site](#)
2. **SkillRC: Is It the Skill or the Stack? Auditing Experience Memory in Agent Harnesses**
Yiwen Lu *Manuscript, 2026* [PDF](#) [Code](#) [Site](#)
3. **SARRA: Attack Resistance Is Only Half the Answer for Search Agents**
Yiwen Lu *Manuscript, 2026* [PDF](#) [Code](#) [Site](#)
4. **Ultra-Wideband Indoor Localization and Bistatic Deep Learning-Based Mapping for ISAC**
Yiwen Lu, Mikhail Saleev, Toros Arikan *Under review at IEEE ICC 2027, SAC – Integrated Sensing and Communications, 2026*
5. **Deep Learning Estimation of the Number and Locations of Reflective Surfaces for Underwater ISAC**
Yiwen Lu, Mikhail Saleev, Yitong Luo, Allison Sproul, Rene Alzina, Toros Arikan *Under review at IEEE ICASSP 2027, 2026* [PDF](#)
6. **Mobility-LLM: Learning Visiting Intentions and Travel Preferences from Human Mobility Data with Large Language Models**
Letian Gong, Yan Lin, Xinyue Zhang, Yiwen Lu, Xuedi Han, Yichen Liu, Shengnan Guo, Youfang Lin, Huaiyu Wan *NeurIPS 2024* [PDF](#)

RESEARCH EXPERIENCE

University of Notre Dame — Independent Researcher 2026 – Present
Reliability and Adaptation in LLM Agent Systems

- Inspired by discussions and encouragement from peers at Notre Dame, pursued independent research on agent reliability, post-training, and experience memory across retrieval and interactive environments.
- **Know2Act**: Pre-registered an eight-variant SFT → GRPO study of knowledge transfer to text-to-SQL agent decisions; agents learned answer structure (89%/82%) without changing their action policies. Found an empty-answer reward issue in 44% of SkyRL tasks and reported it upstream.
- **SkillRC**: Audited experience-memory transfer on ALFWorld and WebShop; showed a +12.7-point gain for GPT-4o-mini but no gain for Qwen3-8B (transport gap −14.8 points), with a pre-registered placebo attributing much of the difference to executor-specific context responses.
- **SARRA**: Evaluated Search-R1/Qwen2.5-3B agent post-training (GRPO, injection, answer supervision) over 75,000 trajectories; answer supervision raised attacked EM from 23.30% to 33.37% and reduced attack-target hits from 21.28% to 2.67%.

University of Notre Dame — Graduate Research Assistant Aug. 2025 – Present
Deep Learning for Spatial Sensing and Environmental Mapping

- **Indoor Localization and Bistatic Mapping**: Investigated ultra-wideband localization and deep-learning-based bistatic environmental mapping for integrated sensing and communications (ISAC); first-author manuscript under review at IEEE ICC 2027.
- **Underwater Scene Reconstruction**: Developed a multi-stage deep-learning and signal-processing pipeline for estimating reflective boundaries from noisy acoustic measurements, incorporating failure diagnosis and correction; first-author manuscript under review at IEEE ICASSP 2027.

Beijing Jiaotong University — Research Assistant Apr. 2023 – Jun. 2025
LLM Adaptation for Spatiotemporal Intelligence / Advisors: Huaiyu Wan and Shengnan Guo

- Co-authored **Mobility-LLM (NeurIPS 2024)**, adapting pretrained LLMs to structured human mobility data for prediction tasks over heterogeneous spatiotemporal sequences.
- Implemented the **Visiting Intention Memory Network (VIMN)** and **Human Travel Preference Prompts (HTPP)**; evaluated the framework across four datasets and three downstream tasks.

TECHNICAL SKILLS

Programming and frameworks: Python, PyTorch, vLLM, Hugging Face Transformers, Search-R1; C/C++, Java, SQL, Bash.

Research methods: LLM post-training (SFT, GRPO), agent harnesses, experience memory, retrieval/tool-use evaluation, pre-registration, paired bootstrap, failure analysis, spatiotemporal and sensor-data modeling.